Lab Assignment 7

Topics:
A. Polymorphism
B. Virtual Function and Pure Virtual Function
C. Operator Overloading

1. Class polygon contains data member width and height and public method set_value() to assign values to width and height. Class Rectangle and Triangle are inherited from polygon class. Both the classes contain public method calculate_area() to calculate the area of Rectangle and Triangle. Use base class pointer to access the derived class object and show the area calculated.

```cpp
#include <iostream>
using namespace std;

class polygon{
    protected:
    float width;
    float height;
    float area;
    public:
    void set_value(float w, float h){
        width=w;
        height=h;
    }
    virtual void calculate_area() {
        area=0;
    }
};
class Rectangle:public polygon{
    public:
    void calculate_area(){
        cout <<"Area of rectangle: "<<height*width<<endl;
    }
};
class Triangle:public polygon{
    public:
    void calculate_area(){
        cout <<"Area of rectangle: "<<(height*width)/2<<endl;
    }
};
```

```cpp
int main(){
    polygon *p;
    Rectangle rect;
    Triangle tri;
    float h;
    float w;
    cout<<"Width: "; cin>>w;
    cout <<"Height: "; cin>>h;
    p=&rect;
    p->set_value(w,h);
    p->calculate_area();
    p=&tri;
    p->set_value(w,h);
    p->calculate_area();
    return 0;
}
```

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shag
Width: 7
Height: 5
Area of rectangle: 35
Area of triangle: 17.5
PS C:\Users\Shagun Garg> cd "c:\Users\Shag
Width: 6
Height: 2
Area of rectangle: 12
Area of triangle: 6
PS C:\Users\Shagun Garg>
```

2. Write a program to create a class shape with functions area and display to find area and display the name of the shape and other essential component of the class. Create derived classes circle, rectangle and triangle each having overridden functions area and display. Write a program to find and display the area of circle, rectangle and triangle.

Users > Shagun Garg > oops1.cpp > Rectangle > Rectangle(float, float)

```cpp
#include <iostream>
using namespace std;
class Shape {
public:
    virtual void area() {
        cout << "Area not defined" << endl;
    }

    virtual void display() {
        cout << "Shape" << endl;
    }
};

class Circle : public Shape {
    float radius;
public:
    Circle(float r) {
        radius = r;
    }
    void area()  {
        cout << "Area of Circle = " << 3.14 * radius * radius << endl;
    }
    void display() {
        cout << "Shape: Circle, Radius = " << radius << endl;
    }
};
```

```cpp
class Rectangle : public Shape {
    float length, breadth;
public:
    Rectangle(float l, float b) {
        length = l;
        breadth = b;
    }
    void area()  {
        cout << "Area of Rectangle = " << length * breadth << endl;
    }
    void display()  {
        cout << "Shape: Rectangle, Length = " << length
             << ", Breadth = " << breadth << endl;
    }
};

class Triangle : public Shape {
    float base, height;
public:
    Triangle(float b, float h) {
        base = b;
        height = h;
    }
    void area() {
        cout << "Area of Triangle = " << 0.5 * base * height << endl;
    }
    void display()  {
        cout << "Shape: Triangle, Base = " << base
             << ", Height = " << height << endl;
    }
};
```

```cpp
int main() {
    Shape *s;
    float r, l, b, tb, th;

    cout << "Enter radius of circle: ";
    cin >> r;
    cout << "Enter length and breadth of rectangle: ";
    cin >> l >> b;
    cout << "Enter base and height of triangle: ";
    cin >> tb >> th;

    Circle c(r);
    Rectangle rect(l, b);
    Triangle t(tb, th);

    s = &c;
    s->display();
    s->area();
    cout << endl;

    s = &rect;
    s->display();
    s->area();
    cout << endl;

    s = &t;
    s->display();
    s->area();

    return 0;
}
```

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Garg\"
Enter radius of circle: 5
Enter length and breadth of rectangle: 2
4
Enter base and height of triangle: 4
5
Shape: Circle, Radius = 5
Area of Circle = 78.5

Shape: Rectangle, Length = 2, Breadth = 4
Area of Rectangle = 8

Shape: Triangle, Base = 4, Height = 5
Area of Triangle = 10
PS C:\Users\Shagun Garg>
```

3. Write a C++ program to compute area of right angle triangle, equilateral triangle, isosceles triangle using function overloading concept.

Users > Shagun Garg > oops2.cpp > main()

```cpp
using namespace std;
#include <iostream>
#include <cmath>
using namespace std;

double area(double base, double height) {
    return 0.5 * base * height;
}
double area(double side) {
    return (sqrt(3) / 4) * pow(side, 2);
}
double area(int equalSide, double base) {
    return 0.25 * base * sqrt(4 * pow(equalSide, 2) - pow(base, 2));
}
int main() {
    double b, h, s, isoBase;
    int isoSide;

    cout << "base and height for Right-Angled Triangle: ";
    cin >> b >> h;
    cout << "Area: " << area(b, h) << endl << endl;

    cout << "side for Equilateral Triangle: ";
    cin >> s;
    cout << "Area: " << area(s) << endl << endl;

    cout << "equal side and base for Isosceles Triangle: ";
    cin >> isoSide >> isoBase;
    cout << "Area: " << area(isoSide, isoBase) << endl;

    return 0;
}
```

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Garg
base and height for Right-Angled Triangle: 6
5
Area: 15.00

side for Equilateral Triangle: 5
Area: 10.83

equal side and base for Isosceles Triangle: 4
6
Area: 7.94
PS C:\Users\Shagun Garg>
```

4. Write a program with Student as abstract class and create derive classes Engineering, Medicine and Science from base class Student. Create the objects of the derived classes and process them and access them using array of pointer of type base class Student.

```
Users > Shagun Garg > oops2.cpp > main()
using namespace std;
#include <iostream>
#include<string>

class Student{
    public:
    virtual void getData() = 0;
    virtual void display() = 0;
};
class Engineering: public Student{
    string name;
    int roll;
    public:
    void getData(){
        cout<<"Name: ";cin>>name;
        cout<<"Roll no.: ";cin>>roll;
    }
    void display(){
        cout << "Engineering Student -> Name: " << name <<endl;
        cout << " Roll no. : " << roll <<endl;
    }
};
class Medicine: public Student{
    string name;
    int roll;
    public:
    void getData(){
        cout<<"Name: ";cin>>name;
        cout<<"Roll no.: ";cin>>roll;
    }
    void display(){
        cout << "Medical Student -> Name: " << name<<endl ;
        cout << " Roll no. : " << roll <<endl;
```

```
};
class Science: public Student{
    string name;
    int roll;
    public:
    void getData(){
        cout<<"Name: ";cin>>name;
        cout<<"Roll no.: ";cin>>roll;
    }
    void display(){
        cout << "Science Student -> Name: " << name<<endl ;
        cout << " Roll no. : " << roll<<endl ;
    }
};
int main(){
    Student *S[3];
    Medicine M;
    Engineering E;
    Science Sc;
    S[0]=&M;
    S[1]=&E;
    S[2]=&Sc;
    for (int i=0;i<3;i++){
        S[i]->getData();
    }
    for (int i=0;i<3;i++){
        S[i]->display();
    }
    return 0;
}
```

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\Users
Name: Shagun
Roll no.: 1024150104
Name: Kiran
Roll no.: 1024150020
Name: Vikas
Roll no.: 1024150028
Medical Student -> Name: Shagun
 Roll no. : 1024150104
Engineering Student -> Name: Kiran
 Roll no. : 1024150020
Science Student -> Name: Vikas
 Roll no. : 1024150028
PS C:\Users\Shagun Garg>
```

5. Create a class Time with three private variables int h,m,s; Create a function to overload '+' operator to add two time variables.

int main(){
Time t1(5,15,34),t2(9,53,58),t3;
t3 = t1 + t2; t3.show();
}

Users > Shagun Garg > oops2.cpp > main()

```cpp
using namespace std;
#include <iostream>

class Time{
    private:
    int h,m,s;
    public:
    Time(int hour=0, int min=0, int sec=0){
        h=hour;
        m=min;
        s=sec;
    }
    Time operator + (Time t){
        return Time(h+t.h,m+t.m,s+t.s);
    }
    void show(){
        cout << h << " hr " << m << " min " << s << " sec" << endl;
    }
};
int main(){
    Time t1(5,15,34),t2(9,53,58),t3;
    t3 = t1 + t2;  t3.show();
    return 0;
}
```

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\
14 hr 68 min 92 sec
PS C:\Users\Shagun Garg>
```

6.Define a class STRING and overload == to compare two strings and + operator for concatenation of two strings.

```
Users > Shagun Garg > oops2.cpp > ...
#include <iostream>
#include <cstring>
using namespace std;

class STRING {
    char str[100];
public:
    STRING() {
        str[0] = '\0';
    }
    void getData() {
        cout << "Enter string: ";
        cin >> str;
    }
    STRING operator + (STRING s) {
        STRING temp;
        strcpy(temp.str, str);
        strcat(temp.str, s.str);
        return temp;
    }
    int operator == (STRING s) {
        if (strcmp(str, s.str) == 0)
            return 1;
        else
            return 0;
    }
    void display() {
        cout << str << endl;
    }
};
```

```
int main() {
    STRING s1, s2, s3;
    cout << "Enter first string:\n";
    s1.getData();
    cout << "Enter second string:\n";
    s2.getData();
    s3 = s1 + s2;
    cout << "\nConcatenated String: ";
    s3.display();
    if (s1 == s2)
        cout << "Strings are equal" << endl;
    else
        cout << "Strings are not equal" << endl;
    return 0;
}
```

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Garg\" ;
Enter first string:
Enter string: Shagun
Enter second string:
Enter string: Shagun

Concatenated String: ShagunShagun
Strings are equal
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Garg\" ;
Enter first string:
Enter string: Shagun
Enter second string:
Enter string: Garg

Concatenated String: ShagunGarg
Strings are not equal
PS C:\Users\Shagun Garg>
```

7.Write a program in C++ to create a class matrix and overload * operator using friend function to multiply two matrices.

8.Overload '[]' to check array index out of bounds problem at run time.

```
Users > Shagun Garg > oops2.cpp > main()
#include <iostream>
using namespace std;

class Array {
    int arr[5];

public:
    void getData() {
        for (int i = 0; i < 5; i++)
            cin >> arr[i];
    }

    int operator[](int index) {
        if (index < 0 || index >= 5) {
            cout << "Index out of bounds!" << endl;
            return -1;
        }
        return arr[index];
    }
};
```

```
int main() {
    Array a;
    a.getData();
    int i;

    while (true) {
        cout << "Enter index : ";
        cin >> i;

        if (i == -1)
            break;

        int value = a[i];
        if (value != -1)
            cout << "Element = " << value << endl;
    }

    return 0;
}
```

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\Users\Sha
2
4
6
8
2
Enter index : 7
Index out of bounds!
Enter index : 9
Index out of bounds!
Enter index : 2
Element = 6
Enter index : 3
Element = 8
Enter index : -1
PS C:\Users\Shagun Garg>
```

9.Overload '()' to input arbitrary number of input arguments for an object.

```
Users > Shagun Garg > oops2.cpp > main()
#include <iostream>
using namespace std;
class Input {
public:
    void operator()(int a) {
        cout << "One value: " << a << endl;
    }

    void operator()(int a, int b) {
        cout << "Two values: " << a << " " << b << endl;
    }

    void operator()(int a, int b, int c) {
        cout << "Three values: " << a << " " << b << " " << c << endl;
    }
```

```
int main() {
    Input obj;
    int choice;
    cout << "Enter number of values : ";
    cin >> choice;

    if (choice == 1) {
        int a;
        cout << "Enter 1 value: ";
        cin >> a;
        obj(a);
    }
    else if (choice == 2) {
        int a, b;
        cout << "Enter 2 values: ";
        cin >> a >> b;
        obj(a, b);
    }
    else if (choice == 3) {
        int a, b, c;
        cout << "Enter 3 values: ";
        cin >> a >> b >> c;
        obj(a, b, c);
    }
    else {
        cout << "Invalid choice!" << endl;
    }

    return 0;
}
```

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\Users\S
Enter number of values : 3
Enter 3 values: 5
6
7
Three values: 5 6 7
PS C:\Users\Shagun Garg> cd "c:\Users\S
Enter number of values : 2
Enter 2 values: 5
1
Two values: 5 1
PS C:\Users\Shagun Garg>
```

10,Write a program in C++ to overload input operator (>>) and output operator (<<).

Users > Shagun Garg > oops2.cpp > operator<<(ostream &, Student &)

```cpp
#include <iostream>
using namespace std;

class Student {
    int roll;
    char name[50];
public:
    friend istream& operator>>(istream &in, Student &s);
    friend ostream& operator<<(ostream &out, Student &s);
};
istream& operator>>(istream &in, Student &s) {
    cout << "Enter roll number: ";
    in >> s.roll;

    cout << "Enter name: ";
    in >> s.name;

    return in;
}
ostream& operator<<(ostream &out, Student &s) {
    out << "Roll Number: " << s.roll << endl;
    out << "Name: " << s.name << endl;

    return out;
}
int main() {
    Student s;

    cin >> s;
    cout << s;

    return 0;
}
```

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagu
Enter roll number: 1024150104
Enter name: Shagun
Roll Number: 1024150104
Name: Shagun
PS C:\Users\Shagun Garg>
```

11.Write a program to convert basic data type (float) to user defined data type (object).

```
class Test {
private:
//… public:
   Test(data_type) {
      // conversion code
   }
};
```

Users > Shagun Garg > oops2.cpp > Test > show()

```cpp
#include <iostream>
using namespace std;
class Test {
    float x;
public:
    Test(float a) {
        x = a;
    }
    void show() {
        cout << "Value = " << x << endl;
    }
};
int main() {
    float y;
    cout << "Enter a float value: ";
    cin >> y;

    Test t = y; // conversion from float to object
    t.show();

    return 0;
}
```

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\
Enter a float value: 5.9999
Value = 5.9999
PS C:\Users\Shagun Garg>
```

12.Write a program to convert UDT to basic data type (float)

```
class Test { public:
   operator data_type() {
      // conversion code
   }
};
```

Users > Shagun Garg > oops2.cpp > ...

```cpp
#include <iostream>
using namespace std;
class Test {
    float x;
public:
    Test(float a) {
        x = a;
    }
    operator float() {
        return x;
    }
};
int main() {
    Test t(5.5);

    float num;
    num = t;
    cout << "Converted value = " << num << endl;

    return 0;
}
```

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\U
Converted value = 5.5
PS C:\Users\Shagun Garg>
```

13.How will you convert one UDT to another UDT. For example conversion of polar to cartesian system.

Polar p(10,5); Cartesian
c = p; c.show();

Users > Shagun Garg > oops2.cpp > Cartesian > x

```cpp
#include <iostream>
#include <cmath>
using namespace std;
class Polar {
public:
    float r, theta;
    Polar(float r1, float t1) {
        r = r1;
        theta = t1;
    }
};
class Cartesian {
    float x, y;
public:
    Cartesian(Polar p) {
        x = p.r * cos(p.theta);
        y = p.r * sin(p.theta);
    }
    void show() {
        cout << "x = " << x << ", y = " << y << endl;
    }
};
int main() {
    float r, theta;
    cout << "Enter radius (r): ";
    cin >> r;
    cout << "Enter angle (theta in radians): ";
    cin >> theta;

    Polar p(r, theta);
    Cartesian c = p;
    c.show();
    return 0;
}
```

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\Users\Sha
Enter radius (r): 6.5
Enter angle (theta in radians): 4.50
x = -1.37017, y = -6.35395
PS C:\Users\Shagun Garg> cd "c:\Users\Sha
Enter radius (r): 2
Enter angle (theta in radians): 4
x = -1.30729, y = -1.5136
PS C:\Users\Shagun Garg>
```